

Sampling & Estimation

ANSWER KEY



Point estimates for the population mean

- 1 A sample of 6 measurements has values 14, 18, 12, 16, 15, 17. Find the unbiased point estimate of the population mean.
$$\bar{x} = \frac{14 + 18 + 12 + 16 + 15 + 17}{6} = \frac{92}{6} \approx 15.333.$$
- 2 Explain why the unbiased estimator of population variance, $s^2 = \frac{1}{n-1} \sum (x_i - \bar{x})^2$, divides by $n - 1$ rather than n .
Dividing by n would systematically underestimate the true population variance on average, since $\bar{x}$ is computed from the same sample, making squared deviations from $\bar{x}$ slightly smaller than deviations from the true (unknown) population mean μ . Dividing by $n - 1$ (the degrees of freedom) corrects this bias.
- 3 A sample of $n = 8$ has $\sum x_i = 152$. Find $\bar{x}$, the unbiased point estimate of μ .
$$\bar{x} = \frac{152}{8} = 19.$$

Computing the unbiased sample variance

- 4 For the sample 5, 8, 10, 7, 10, find the unbiased estimate of the population variance.
Mean = $\frac{5 + 8 + 10 + 7 + 10}{5} = 8$. Squared deviations: 9, 0, 4, 1, 4, summing to 18. $s^2 = \frac{18}{5-1} = 4.5$.
- 5 A sample of $n = 10$ has $\sum (x_i - \bar{x})^2 = 81$. Find the unbiased estimate of the population variance and standard deviation, to three decimal places.
 $s^2 = \frac{81}{10-1} = 9$. $s = \sqrt{9} = 3.000$.
- 6 For the sample 12, 9, 14, 11, 13, 15, find the unbiased estimate of the population standard deviation, to three decimal places.
Mean = $\frac{12 + 9 + 14 + 11 + 13 + 15}{6} = \frac{74}{6} \approx 12.333$. Sum of squared deviations
 $\approx 0.111 + 11.111 + 2.778 + 1.778 + 0.444 + 7.111 = 23.333$. $s^2 = \frac{23.333}{5} \approx 4.667$.
 $s = \sqrt{4.667} \approx 2.160$.

The Central Limit Theorem

- 7 State the Central Limit Theorem as it applies to the sample mean $\bar{X}$ of a random sample of size n from a population with mean μ and variance σ^2 .
For sufficiently large n , $\bar{X}$ is approximately normally distributed with mean μ and variance $\frac{\sigma^2}{n}$, regardless of the shape of the original population distribution — this holds even if the underlying population is not normal, provided n is large enough (a common guideline is $n \geq 30$).

- 8 A population has mean $\mu = 40$ and standard deviation $\sigma = 12$. A random sample of size $n = 36$ is taken. Using the Central Limit Theorem, state the approximate distribution of $\bar{X}$.
 $\bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right)$ approximately, with $\frac{\sigma^2}{n} = \frac{144}{36} = 4$. So $\bar{X} \sim N(40, 4)$ approximately, i.e. mean 40, standard deviation 2.
- 9 Explain why the Central Limit Theorem is so useful in practice, even when the underlying population distribution is unknown or non-normal.
- It allows normal-distribution-based methods (confidence intervals and hypothesis tests for μ) to be used even without knowing the population's exact distribution shape, as long as the sample size is reasonably large — making inference about μ possible in a very wide range of real-world situations.

Using the distribution of the sample mean

- 10 Using $\bar{X} \sim N(40, 4)$ from above, find $P(\bar{X} > 43)$, given $\Phi(1.5) \approx 0.9332$.
 $z = \frac{43 - 40}{2} = 1.5$. $P(\bar{X} > 43) = 1 - \Phi(1.5) = 1 - 0.9332 = 0.0668$.
- 11 Using the same distribution, find $P(38 < \bar{X} < 41)$, given $\Phi(0.5) \approx 0.6915$ and $\Phi(-1) \approx 0.1587$.
 $z_1 = \frac{38 - 40}{2} = -1$, $z_2 = \frac{41 - 40}{2} = 0.5$.
 $P(38 < \bar{X} < 41) = \Phi(0.5) - \Phi(-1) = 0.6915 - 0.1587 = 0.5328$.

Confidence intervals for μ with known σ

- 12 A sample of $n = 49$ has mean $\bar{x} = 68$ and known population standard deviation $\sigma = 7$. Construct a 95% confidence interval for μ , using $z^* = 1.96$.
Standard error: $\frac{7}{\sqrt{49}} = 1$. Margin of error: $1.96(1) = 1.96$. Confidence interval: 68 ± 1.96 , or $(66.04, 69.96)$.
- 13 A sample of $n = 64$ has mean $\bar{x} = 25$ and $\sigma = 8$. Construct a 90% confidence interval for μ , using $z^* = 1.645$.
Standard error: $\frac{8}{\sqrt{64}} = 1$. Margin of error: $1.645(1) = 1.645$. Confidence interval: 25 ± 1.645 , or $(23.355, 26.645)$.

Confidence intervals with unknown σ (large samples)

- 14 A sample of $n = 100$ has mean $\bar{x} = 52.4$ and sample standard deviation $s = 9$. Since n is large, construct a 99% confidence interval for μ using $z^* = 2.576$ as an approximation, to two decimal places.
Standard error: $\frac{9}{\sqrt{100}} = 0.9$. Margin of error: $2.576(0.9) \approx 2.3184$. Confidence interval: 52.4 ± 2.32 , or approximately $(50.08, 54.72)$.

- 15 A sample of $n = 50$ has mean $\bar{x} = 110$ and sample standard deviation $s = 15$. Construct a 90% confidence interval for μ , using $z^* = 1.645$, to two decimal places.
Standard error: $\frac{15}{\sqrt{50}} \approx 2.1213$. Margin of error: $1.645(2.1213) \approx 3.49$. Confidence interval: 110 ± 3.49 , or approximately (106.51, 113.49).
-

Determining the required sample size

- 16 A researcher wants a 95% confidence interval for μ with margin of error at most 1.5, given $\sigma = 10$. Find the minimum sample size, using $z^* = 1.96$, rounding up.
$$n = \left(\frac{z^* \sigma}{E} \right)^2 = \left(\frac{1.96(10)}{1.5} \right)^2 \approx (13.067)^2 \approx 170.74$$
. Rounding up: $n = 171$.
- 17 A researcher wants a 99% confidence interval for μ with margin of error at most 3, given $\sigma = 15$. Find the minimum sample size, using $z^* = 2.576$, rounding up.
$$n = \left(\frac{2.576(15)}{3} \right)^2 = (12.88)^2 \approx 165.89$$
. Rounding up: $n = 166$.
-

Interpreting confidence intervals

- 18 A 95% confidence interval for a population mean is (41.2, 48.8). State the correct interpretation of this interval.
If this sampling and interval-construction process were repeated many times, approximately 95% of the resulting intervals would contain the true population mean. It is NOT correct to say there is a 95% probability the true mean lies in this specific interval, since the true mean is a fixed (though unknown) value, not a random variable.
- 19 Explain why increasing the confidence level from 90% to 99%, with the sample data unchanged, produces a wider confidence interval.
A higher confidence level requires a larger critical value z^* (approximately 2.576 for 99% versus 1.645 for 90%). Since the margin of error is directly proportional to z^* , a larger z^* produces a wider interval — there is an inherent tradeoff between confidence level and precision.
-

A well-designed word problem: estimating average commute time

- 20** This question has four parts. A researcher surveys $n = 64$ randomly selected commuters, finding sample mean commute time $\bar{x} = 34.5$ minutes with sample standard deviation $s = 12$ minutes. (a) State the unbiased point estimate of the population mean commute time. (b) Since n is large, construct a 95% confidence interval for the true mean, using $z^* = 1.96$. (c) The city claims the average commute is 30 minutes. Using your confidence interval, comment on whether this claim is plausible. (d) The researcher wants a margin of error of at most 2 minutes at the same confidence level. Find the minimum required sample size, using $s = 12$ as an estimate of σ , rounding up.

(a) 34.5 minutes. (b) Standard error: $\frac{12}{\sqrt{64}} = 1.5$. Margin of error: $1.96(1.5) = 2.94$. Confidence interval: 34.5 ± 2.94 , or (31.56, 37.44). (c) Since 30 minutes lies outside this interval (below the lower bound of 31.56), there is evidence at the 95% confidence level that the true mean commute time exceeds the city's claimed 30 minutes — the claim is not well supported by the data. (d) $n = \left(\frac{1.96(12)}{2} \right)^2 = (11.76)^2 \approx 138.30$. Rounding up: $n = 139$.